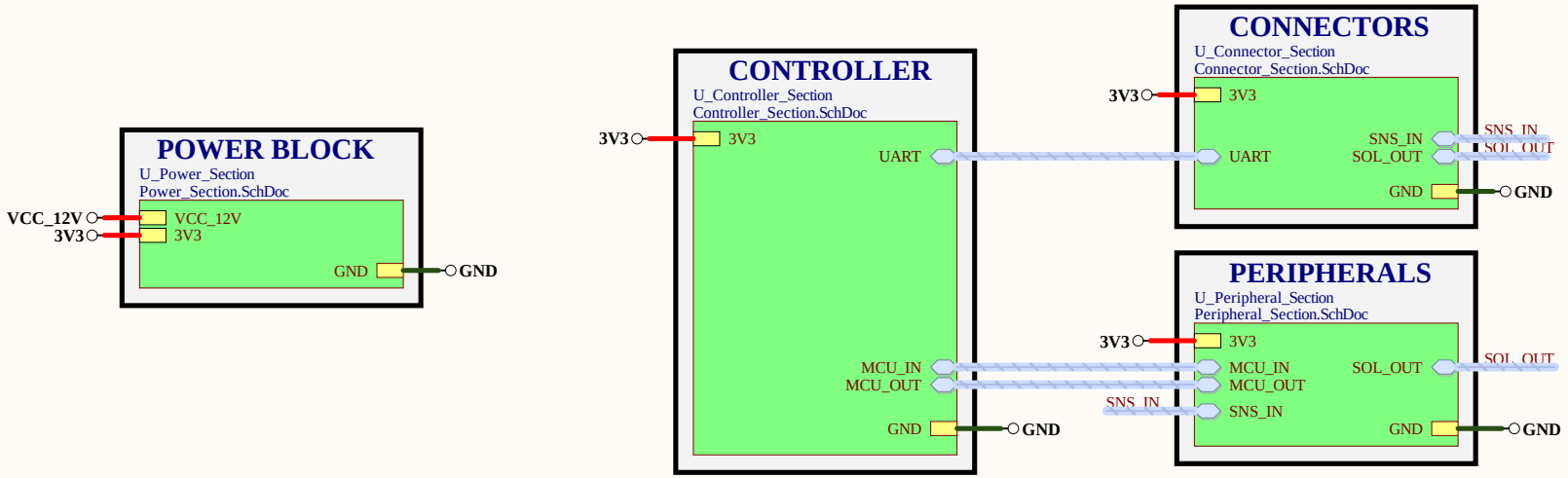
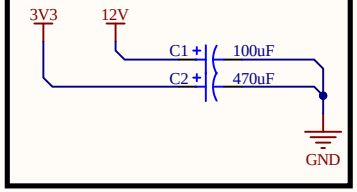
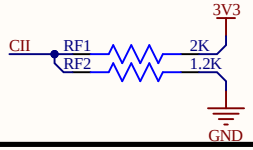

POWER BLOCK
U_Power_Section
Power_Section.SchDoc
VCC_12V
3V3
VCC_12V
3V3
GND
GND
CONTROLLER
U_Controller_Section
Controller_Section.SchDoc
3V3
3V3
UART
MCU_IN
MCU_OUT
GND
GND
CONNECTORS
U_Connector_Section
Connector_Section.SchDoc
3V3
3V3
UART
SNS_IN
SOL_OUT
SNS_IN
SOL_OUT
GND
GND
PERIPHERALS
U_Peripheral_Section
Peripheral_Section.SchDoc
3V3
3V3
MCU_IN
MCU_OUT
SNS_IN
SOL_OUT
SOL_OUT
GND
GND

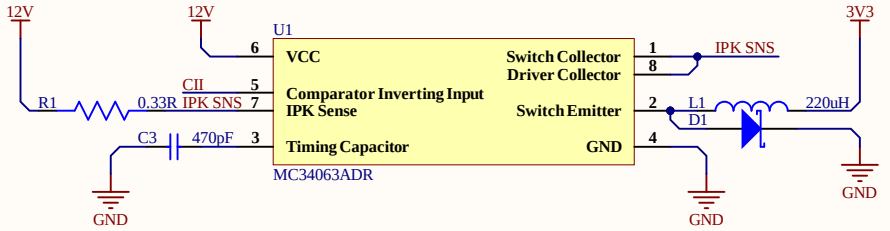
## DECOUPLING CAPS

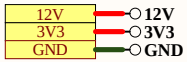



## FEEDBACK NETWORKS




## 12-3.3V BUCK CONVERTER

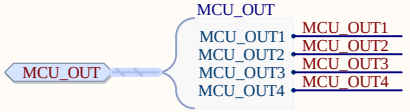


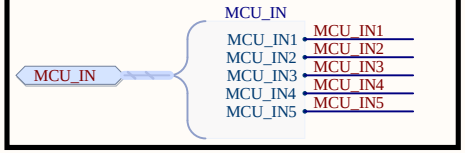
## UART SIGNALS



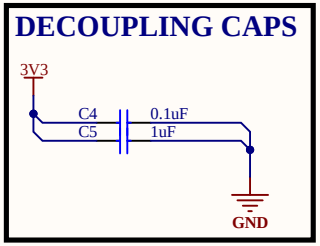

## SOLENOID CONTROL SIGNAL



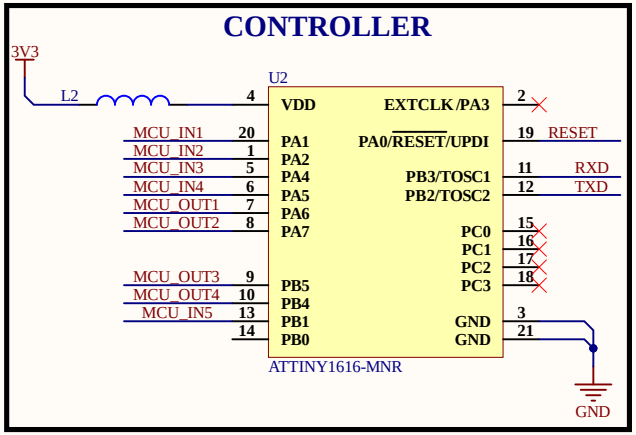

## CONTROLLER SENSE INPUT



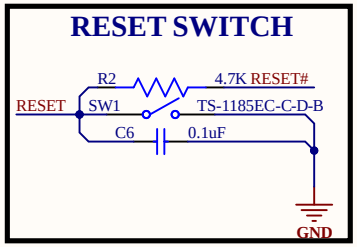

## DECOUPLING CAPS




## CONTROLLER

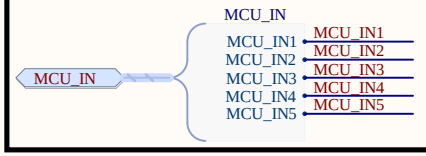



## RESET SWITCH

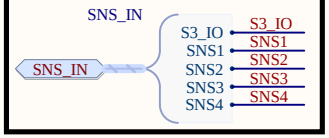




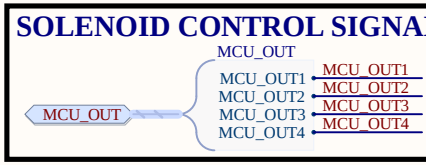

## CONTROLLER SENSE INPUT



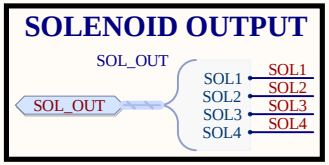

## LEVEL SENSE INPUT



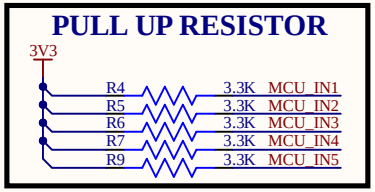

## SOLENOID CONTROL SIGNAL



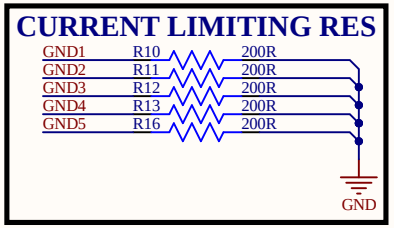

## SOLENOID OUTPUT




## PULL UP RESISTOR



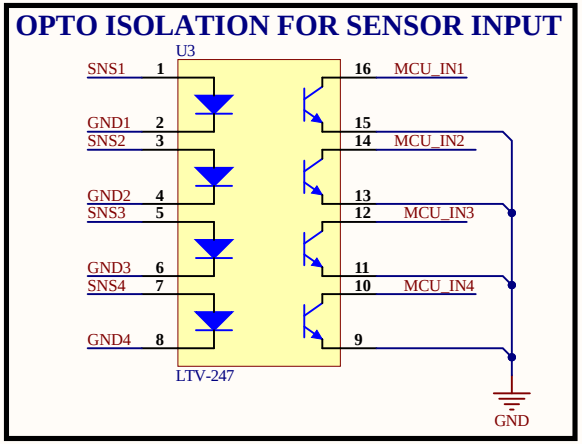

## CURRENT LIMITING RES



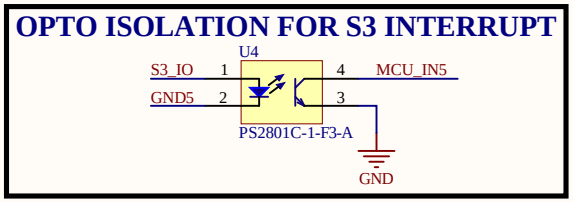




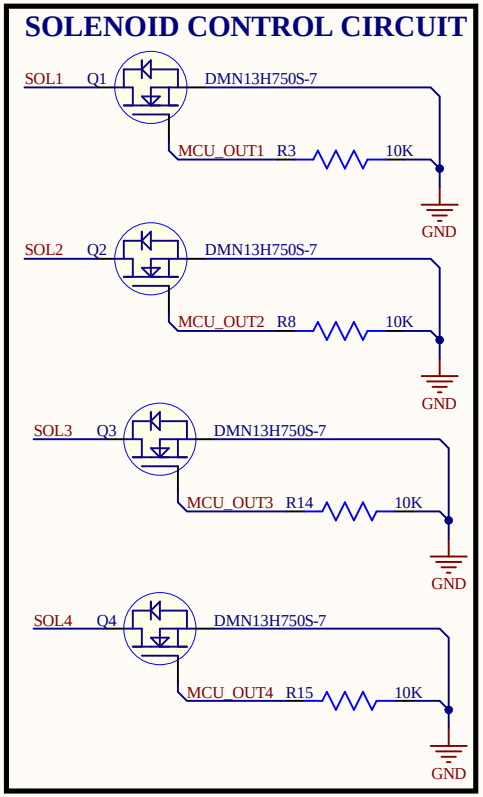
## OPTO ISOLATION FOR SENSOR INPUT




## OPTO ISOLATION FOR S3 INTERRUPT



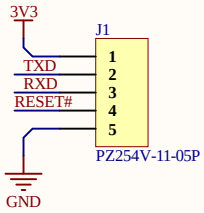

## SOLENOID CONTROL CIRCUIT



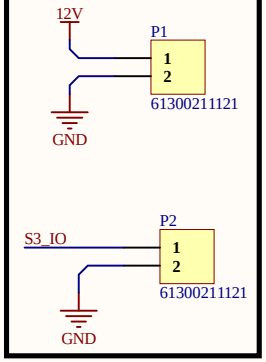
## UART SIGNALS



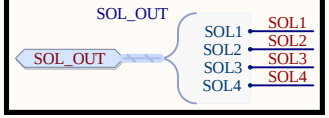

## PROGRAMMING HEADER



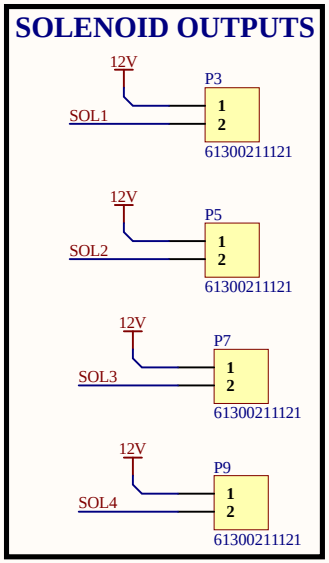

## S3 SIGNALS



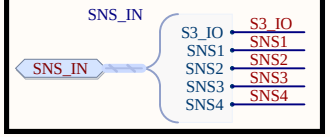

## SOLENOID OUTPUT



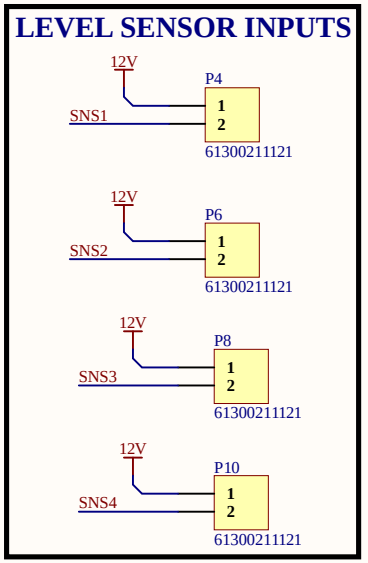

## SOLENOID OUTPUTS

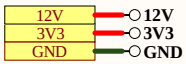


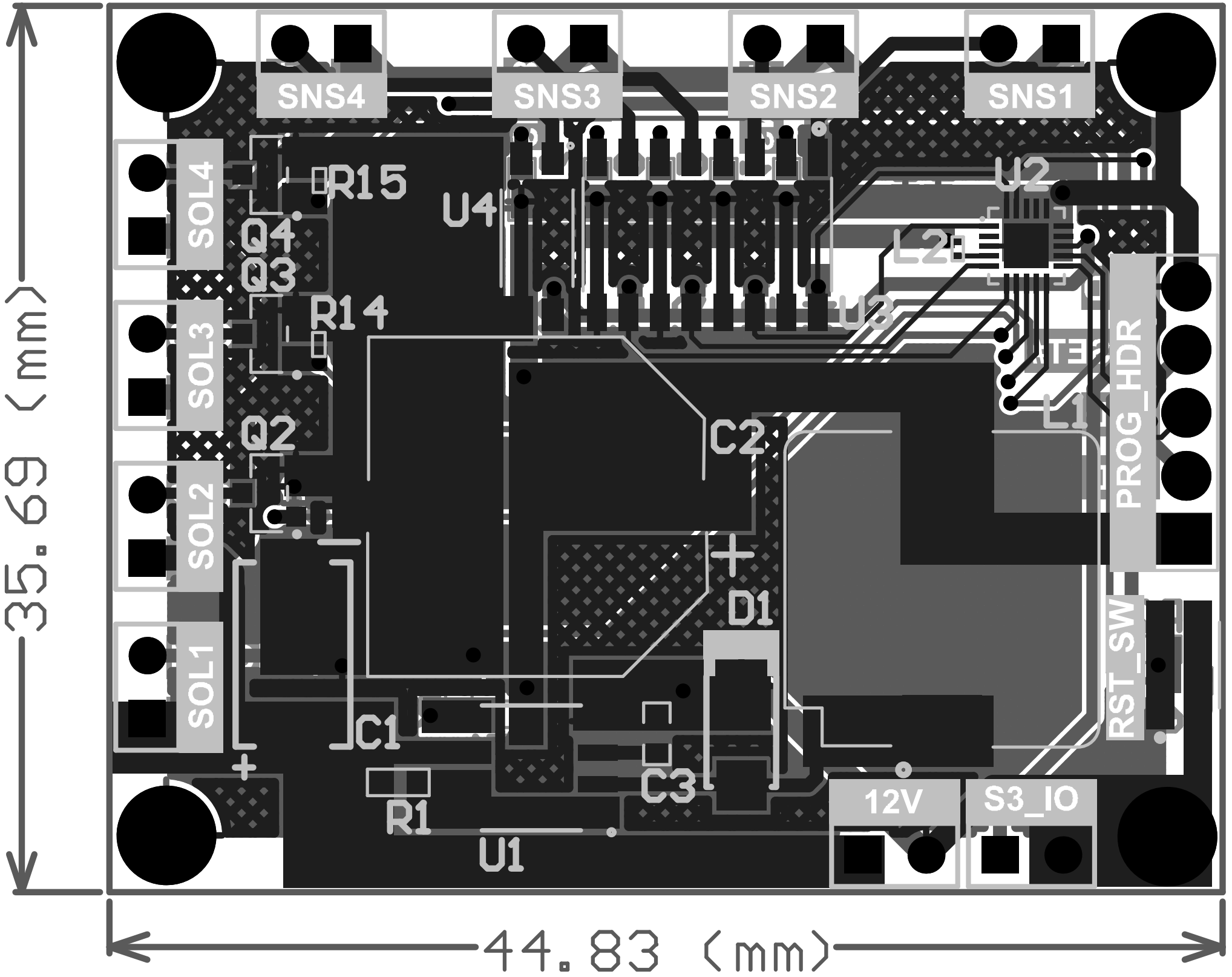

## LEVEL SENSE INPUT




## LEVEL SENSOR INPUTS




SNS4
SNS3
SNS2
SNS1
SOL4
SOL3
SOL2
SOL1
R15
U4
Q4
Q3
R14
U2
L2
U3
L1
PROG_HDR
Q2
C2
D1
RST_SW
C1
C3
R1
U1
12V
S3_IO
35.69 (mm)
44.83 (mm)

| Comment | Description | Designator | Footprint | LibRef | Quantity |
|---|---|---|---|---|---|
| 100uF | CAP TANT 100UF 20% 25V 2917 | C1 | CAP- 2917 | TAJE107M025RNJ | 1 |
| 470uF | 470uF ±20% 50V 12.5mm 13.5mm 1000hrs 105°C -40°C~+105°C SMD,12.5x13.5mm Aluminum Electrolytic Capacitors - SMD RoHS | C2 | VT1H471MI135000CE0 | VT1H471MI135000CE0 | 1 |
| 470pF | CAP CER 470pF 10% X7R 50V 0603 SMD | C3 | Cap0603 | CC0603KRX7R9BB471 | 1 |
| 0.1uF | CAP CER 0.1UF 50V Y5V 0603 | C4, C6 | Cap0603 | CC0603ZRY5V9BB104 | 2 |
| 1uF | CAP CER 1UF 25V X5R 0603 | C5 | Cap0603 | CC0603KRX5R8BB105 | 1 |
| SS24A | 40V 500mV@2A 2A SMA(DO-214AC) Schottky Barrier Diodes (SBD) ROHS | D1 | SMA | SS24A | 1 |
| PZ254V-11-05P | Pin Header 5 1 right-angle | J1 | HDR1X5 | PZ254V-11-05P | 1 |
| 220uH | Inductor Power Shielded Wirewound 220uH 20% 1KHz Ferrite 1.3A 310mOhm DCR Blister T/R | L1 | SLF12575T-221M1R3-PF | SLF12575T-221M1R3-PF | 1 |
| MMZ0603S601CT000 | FERRITE BEAD 600 OHM 0201 1LN | L2 | IND 0201 | MMZ0603S601CT000 | 1 |
| 61300211121 | CONN HEADER VERT 2POS 2.54MM | P1, P2, P3, P4, P5, P6, P7, P8, P9, P10 | HDR1X2 | Header 1x2 | 10 |
| DMN13H750S-7 | MOSFET N-CH 130V 1A SOT23 | Q1, Q2, Q3, Q4 | SOT23-3 | DMN13H750S-7 | 4 |
| 0.33R | 0.33Ω 0603 Chip Resistor - Surface Mount ROHS | R1 | RES-0603 | RTT03R330FTP | 1 |
| 4.7K | RES SMD 4.7K OHM 1% 1/10W 0603 | R2 | RES-0603 | RC0603FR-074K7L | 1 |
| 10K | RES SMD 10K OHM 1% 1/20W 0201 | R3, R8, R14, R15 | Res-0201 | RC0201FR-0710KL | 4 |
| 3.3K | RES SMD 3.3K OHM 1% 1/10W 0603 | R4, R5, R6, R7, R9 | RES-0603 | RC0603FR-073K3L | 5 |
| 200R | RES SMD 200 OHM 0.5% 1/10W 0603 | R10, R11, R12, R13, R16 | RES-0603 | RT0603DRD07200RL | 5 |
| 2K | RES SMD 2K OHM 1% 1/10W 0603 | RF1 | RES-0603 | RC0603FR-072KL | 1 |
| 1.2K | ±1% 1/10W Thick Film Resistors 75V ±100ppm/°C -55°C~+155°C 1.2kΩ 0603 Chip Resistor - Surface Mount ROHS | RF2 | RES-0603 | 0603WAF1201T5E | 1 |
| TS-1185EC-C-D-B | 3mm 2mm Round Button 50mA Top3.65mm SPST 12V SMD Tactile Switches | SW1 | TS-1185EC-C-D-B | TS-1185EC-C-D-B | 1 |
| MC34063ADR | 1.5-A boost, buck, inverting switching regulator, operation from 0°C to 70°C 8-SOIC 0 to 70 | U1 | SOIC-8 | MC34063ADR | 1 |
| ATTINY1616-MNR | IC MCU 8BIT 16KB FLASH 20VQFN | U2 | 20-QFN | ATTINY1616-MNR | 1 |
| LTV-247 | OPTOISO 3.75KV 4CH TRANS 16SO | U3 | LTV-247 | LTV-247 | 1 |
| PS2801C-1-F3-A | TransistorOptocouplers DC 1 2500Vrms SOP-4_P1.27 Optocouplers RoHS | U4 | PS2801C-1-F3-A | PS2801C-1-F3-A | 1 |